THE BASE CHANGE MAP FOR THE SHEAF WITH EXACT SUPPORT IS AN ISOMORPHISM FOR A FLAT MORPHISM AND A PSEUDO-COHERENT CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E2M (sources/stacks-duality/pinned-original-source.tex line 2595).

Allowed standard prerequisites (editorial).

- The base change map for the sheaf with exact support and the criterion for its being an isomorphism. Same chapter, equation labelled equation-base-change-exact-support, invoked in the statement, and Lemma labelled lemma-check-base-change-is-iso, cited at proof lines 2605-2608. Fixes the arrow and reduces the affine case to an algebraic equality.
- Comparison of derived Hom between rings and schemes for a pseudo-coherent module. Derived Categories of Schemes, Lemma labelled perfect-lemma-quasi-coherence-internal-hom, cited at proof lines 2608-2614. Justifies checking the equality of ring level Homs.
- The tensor evaluation isomorphism in the flat case. More on Algebra, Lemma labelled more-algebra-lemma-internal-hom-evaluate-tensor-isomorphism part (3), cited at proof lines 2616-2618, with Dualizing Complexes, Section labelled dualizing-section-base-change-trivial-duality. Supplies the displayed equality.
- The stalk level description of the base change map and its surjectivity criterion. Dualizing Complexes, Equation labelled dualizing-equation-base-change and Lemma labelled dualizing-lemma-flat-bc-surjection, cited at proof lines 2624-2638. Give the second proof.

Source reference. Original equation 0E2K: sources/stacks-duality/pinned-original-source.tex, source lines 2533–2533. Same-chapter pointer into the author environment equation at source line 2533; invoked at proof line 2627 of the counted proof.

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Source reference. Original lemma 0E2L: sources/stacks-duality/pinned-original-source.tex, source lines 2561–2561. Same-chapter pointer into the author environment lemma at source line 2561; invoked at proof line 2605 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A6H: sources/stacks-chow/tags.txt, source lines 12845–12845. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2613 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATK: sources/stacks-chow/tags.txt, source lines 13618–13618. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2619 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0E28: sources/stacks-chow/tags.txt, source lines 17598–17598. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2621 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0E29: sources/stacks-chow/tags.txt, source lines 17599–17599. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2636 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BZM: sources/stacks-chow/tags.txt, source lines 15055–15055. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2638 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In the situation above, assume f is flat and i pseudo-coherent. Then (0E2K) is an isomorphism for K in $D^+_{QCoh}(\mathcal{O}_X)$.*

Proof. First proof. To prove this map is an isomorphism, we may work locally. Hence we may assume X, X', Z, Z' are affine, say corresponding to the rings A, A', B, B'. Then B and A' are tor independent over A. By Lemma 0E2L it suffices to check that

$$R\operatorname{Hom}_A(B, K) \otimes^{\mathbf{L}}_A A' = R\operatorname{Hom}_{A'}(B', K \otimes^{\mathbf{L}}_A A')$$

in $D(A')$ for all $K \in D^+(A)$. Here we use Derived Categories of Schemes, Lemma 0A6H and the fact that B, resp. B' is pseudo-coherent as an A-module, resp. A'-module to compare derived hom on the level of rings and schemes. The displayed equality follows from More on Algebra, Lemma 0ATK part (3). See also the discussion in Dualizing Complexes, Section 0E28.

Second proof[1]. Let $z' \in Z'$ with image $z \in Z$. First show that (0E2K) on stalks at z' induces the map

$$R\operatorname{Hom}(\mathcal{O}_{Z,z}, K_z) \otimes^{\mathbf{L}}_{\mathcal{O}_{Z,x}} \mathcal{O}_{Z',z'} \longrightarrow R\operatorname{Hom}(\mathcal{O}_{Z',z'}, K_z \otimes^{\mathbf{L}}_{\mathcal{O}_{X,z}} \mathcal{O}_{X',z'})$$

from Dualizing Complexes, Equation (0E29). Namely, the constructions of these maps are identical. Then apply Dualizing Complexes, Lemma 0BZM. □

[1]This proof shows it suffices to assume K is in $D^+(\mathcal{O}_X)$.